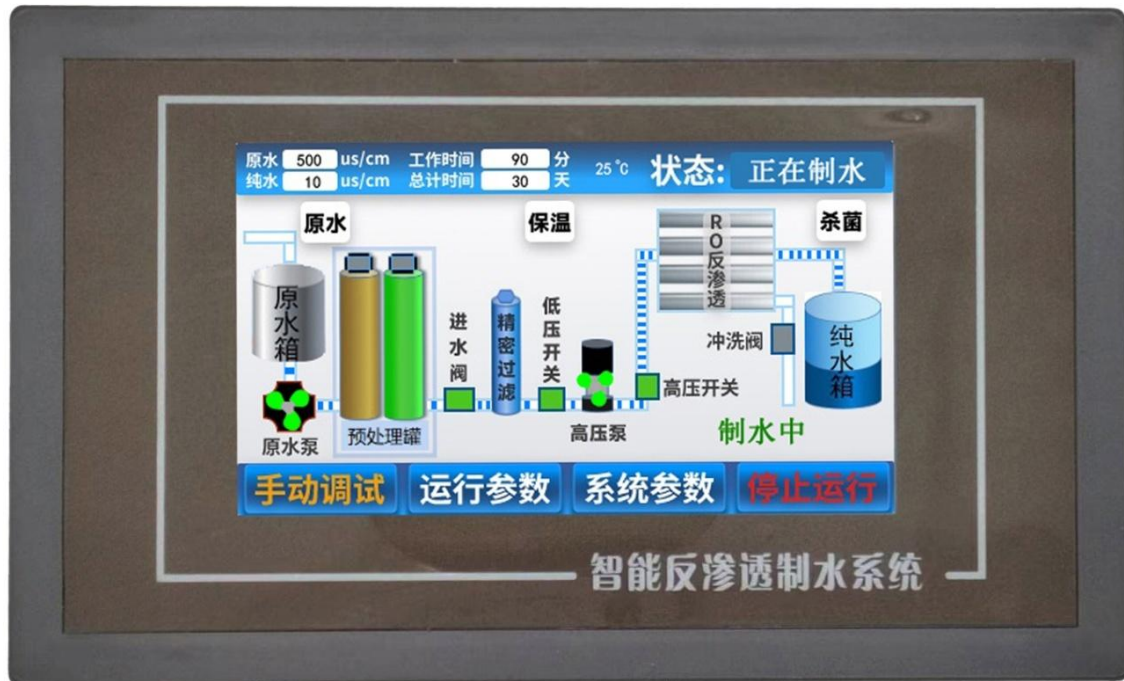
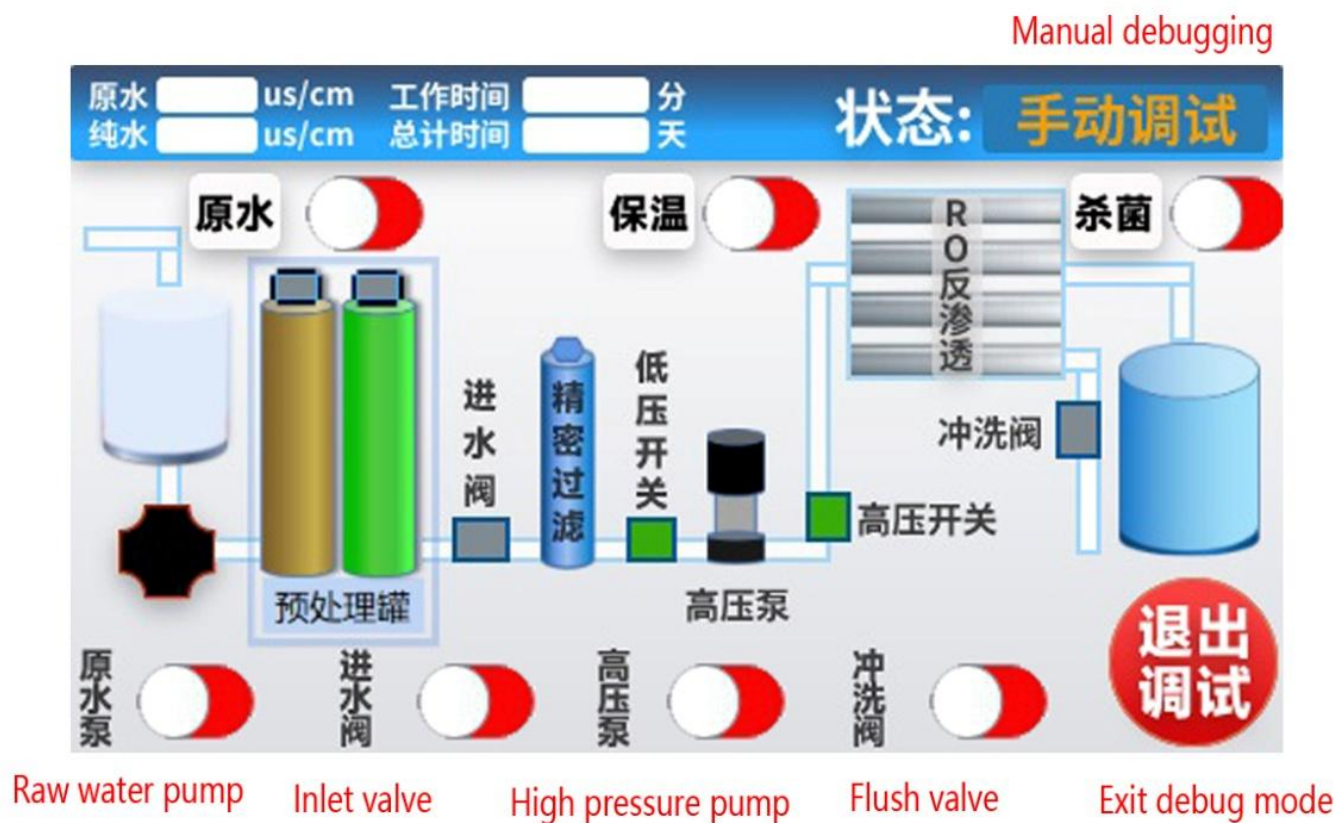
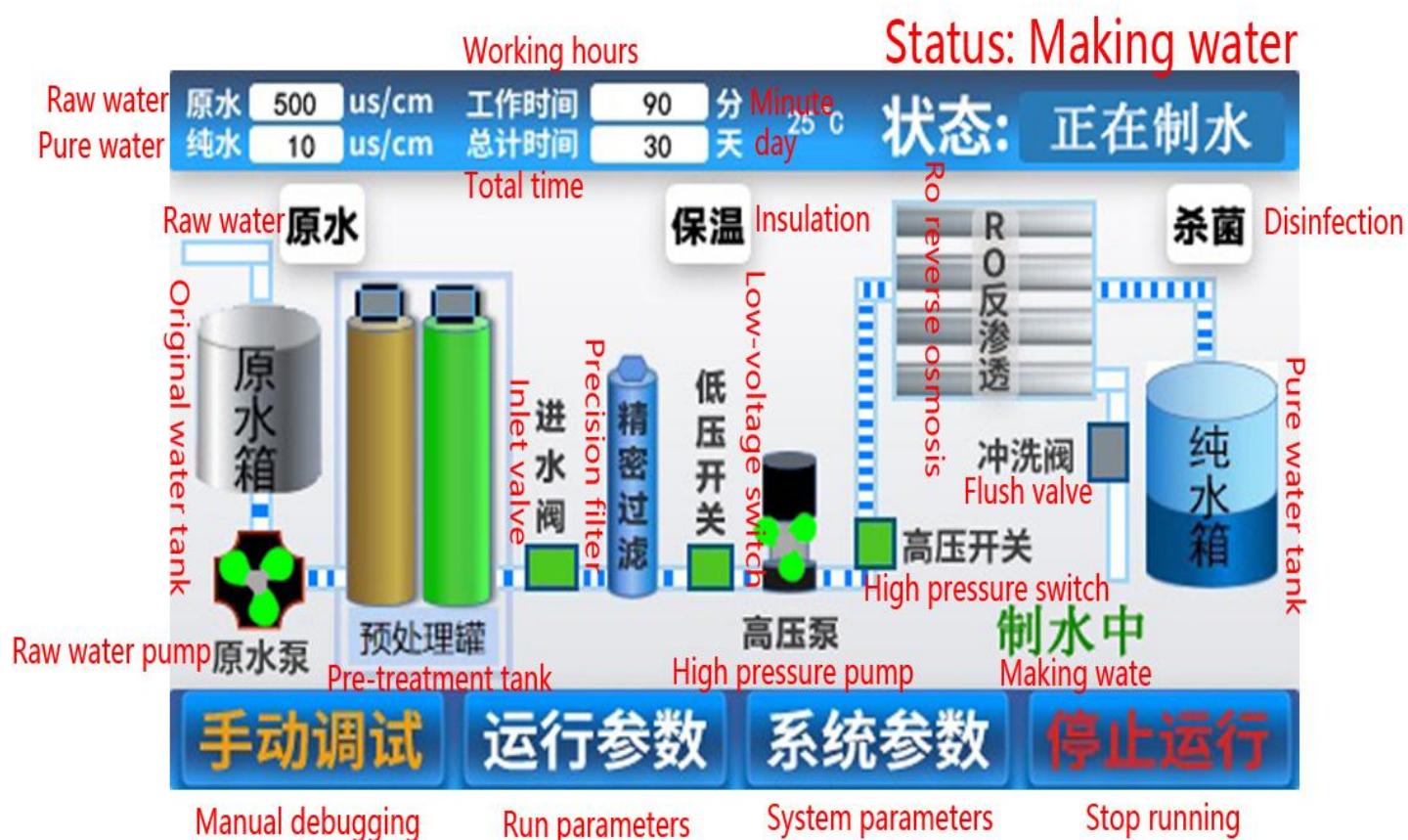


HK-R520 Reverse osmosis control system operation manual



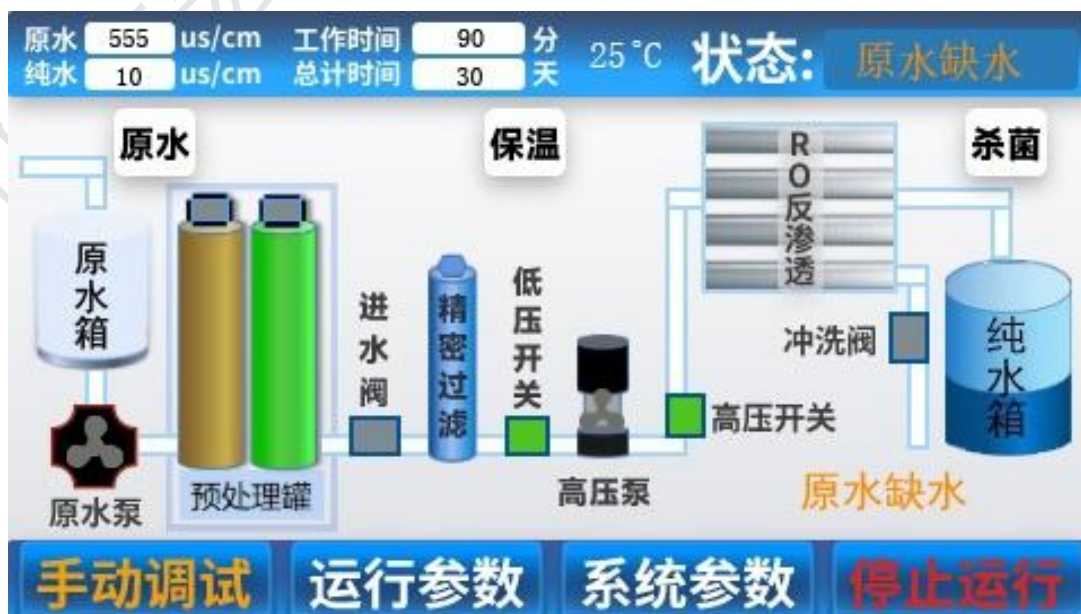
R520 Single-stage reverse osmosis controller with a 4.3-inch color high-definition LCD touch screen, the interface dynamically displays the system and the machine combined, fault issues, and operation steps are clearly visible. The wiring is equipped with pluggable terminals, which are anti-reversal and anti-loose connection, maintenance convenient. The card slot fixing machine is easy to install, and the accessory wiring position is clearly marked and easy to understand. It can be extended to network and remote control, and the manufacturer information can be customized.



Functional introduction

1. Dry protection

Monitoring of the water source used as a source of water, when the raw water is interrupted, the low water level of the storage tank, to prevent the system idling, the controller automatically stops the operation of the entire system, LCD display status prompt: (Original water shortage) **原水缺水**, The entire control system stops running. The controller constantly monitors the no-water switch, and when the liquid level returns to normal, the controller automatically restores system operation



2、 Low voltage protection

When the pretreatment system is in a state of water shortage or the low pressure switch is damaged, and cannot provide the RO system with normal water supply pressure (ins low pressure), the controller temporarily shuts down the water production system. LCD display status prompt: (Low pressure anomaly)

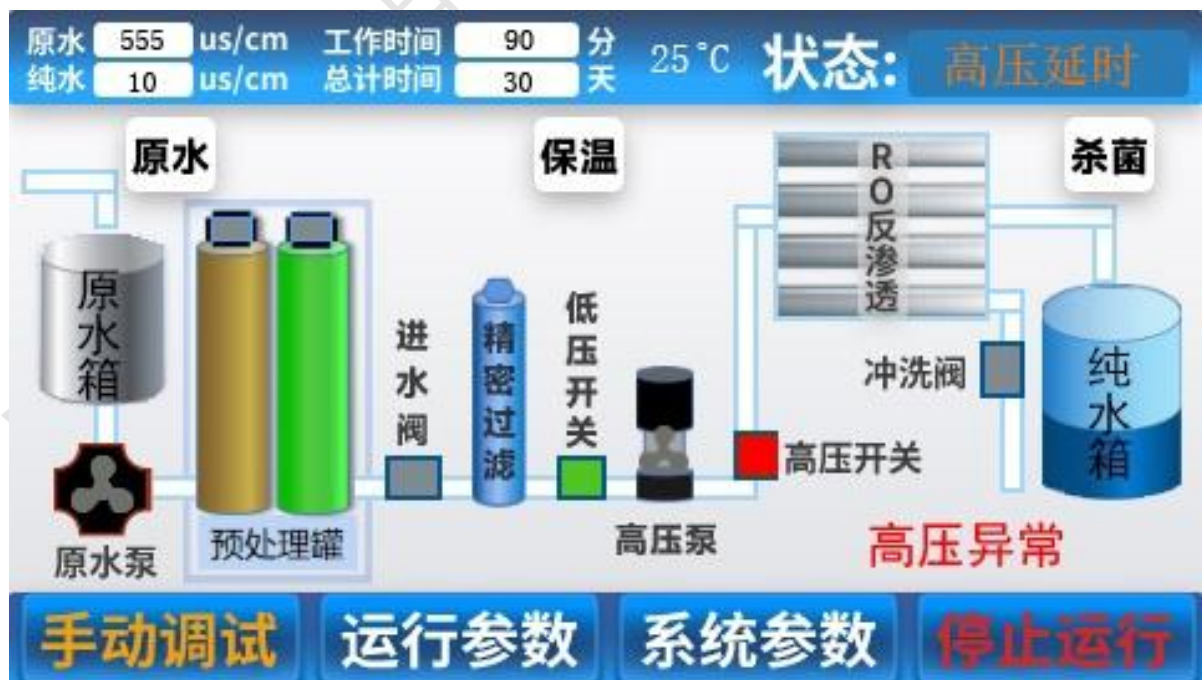
低压异常， Display interface stops dynamic operation, fault display at low pressure switch: **red**。

Thereafter, the controller will continuously detect the low pressure switch, and when the pressure returns to normal, the water control system will operate normally after reaching the set time



3、High voltage protection

When the pressure in front of the RO membrane is too high or the high-pressure switch is damaged, the controller shuts down the entire RO system, LCD display status prompt: (High voltage anomaly) 高压异常, High pressure switch fault display: red. After that, the controller will detect the high-pressure switch without interruption, and the water production control system will operate normally when the pressure returns to normal and the time is reached.



4、Membrane flushing function

When the protection device and status are operating normally, the system performs a first membrane flush each time it is turned on for operation, and during water production, the defaults to a flush of 15 seconds every 60 minutes (this parameter can be manually adjusted). After the pure water tank is full, a membrane flush is performed, followed by a flush of 15 seconds every 2 hours in standby mode (this parameter can be manually adjusted).



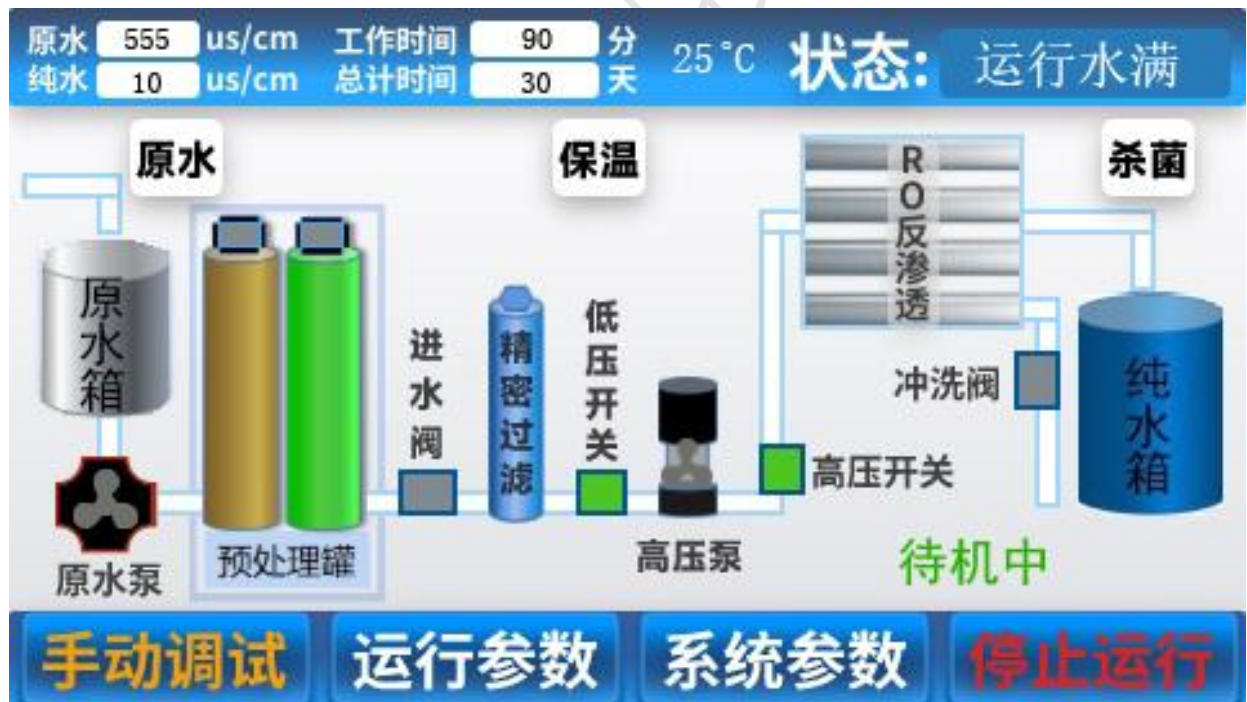
5、Feed water tank level control function

When the water level of the raw water tank is lower than the high-level float ball, the control system opens the raw water valve to fill water, and when water level of the raw water tank reaches the high water level, the control system will close the raw water valve.



6、Pure water tank level control function

When the level of the pure water tank is lower than the high-level float ball, the control system will start the water production system to produce water, until the of the pure water tank reaches the preset high level, the system completes the membrane flush and automatically enters the standby state.

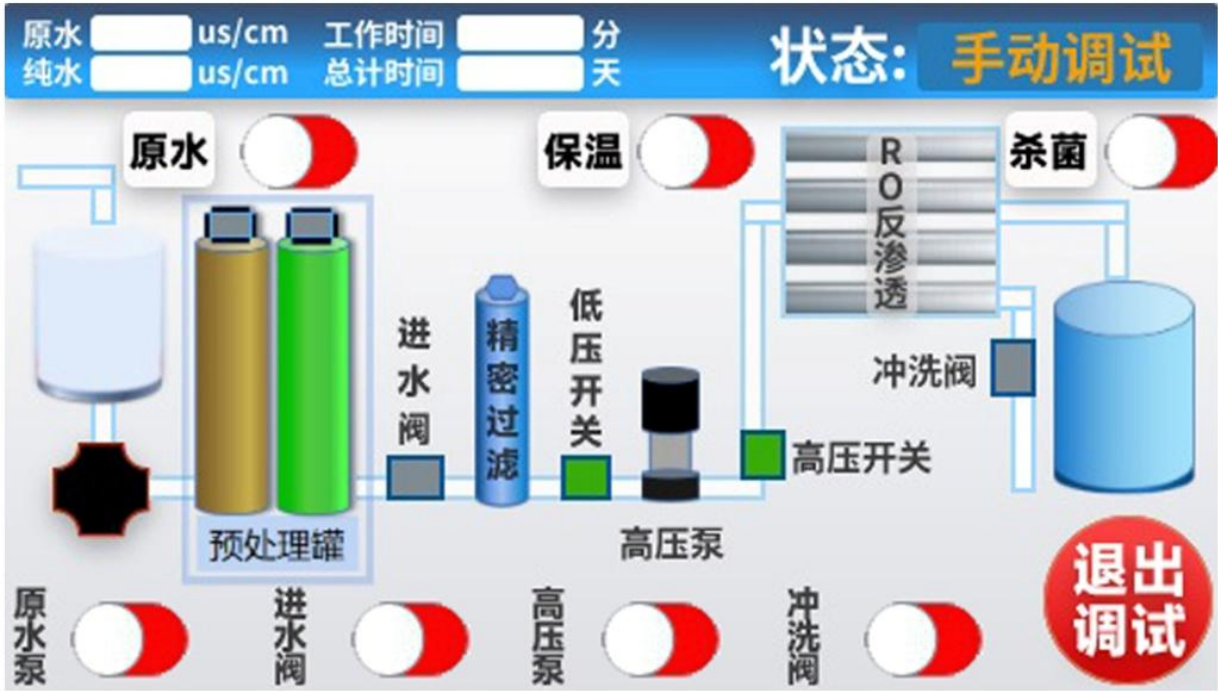


7、 Manual tuning function

Test if each output is normal, the button is in  state after manual start.

Note: When manual tuning is enabled, please press the buttons in order.

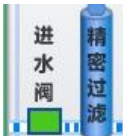
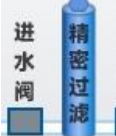
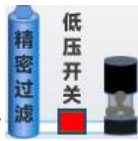
Manual debugging



Raw water pump Inlet valve High pressure pump Flush valve Exit debug mode

8、 Screen on indicator and off state

Original water tank valve operation indicator	Raw water valve  open	Raw water valve closed 
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Raw water pump operation indication	Raw water pump startup 	Raw water pump shutdown 
Valve head flush indication	Valve head flush start 	Valve head flush closed 
Inlet valve operation indication	Inlet valve start 	Inlet valve closed 
Low-voltage switch indicator	Low pressure normal 	Low pressure anomaly 
High pressure pump operation indicator	High pressure pump start 	High pressure pump off 
High pressure switch indicator	High pressure normal 	High voltage anomaly 
Flush valve operation indicator	Flush valve switch start 	Flush valve switch closed 
Ozone sterilization	Ozone sterilization	Ozone sterilization

operation indicator	 start	 off
Temperature control heating operation indicator	Temperature control heating start 	Temperature control heating of
Source water quality testing		Detect the concentration of heavy metal ions in the raw water tank.
Pure water quality testing		Detect the concentration of heavy metal ions in the pure water tank.

Product Parameters

1. Power supply voltage: DC24V;
2. Display: 4.3-inch LCD screen;
3. Touch type: Resistive touch screen;
4. Backlight life: >20000 hours (time to half brightness when operating continuously at maximum brightness);
5. Consumption power: $\geq 5W$;
6. Working environment: Temperature $-20-70^{\circ}C$; Relative humidity $\leq 85\% RH$;

- 7. Water quality detection: high precision dual-channel conductivity;
- 8. Case size: 165mm*100mm*43mm;

Run parameters setting

- 1. Tap the **运行参数** Run Parameters button on the main interface to enter the settings interface;
- 2. The range and description of the running parameters are shown in the following figure;

运行冲洗时间 : 15 秒	运行冲洗间隔 : 60 分	
Run time flush	Run wash interval	
待机冲洗时间 : 15 秒	待机冲洗间隔 : 2 时	
Standby flush time	Standby flush interval	
冲洗方式 : 高压冲洗 ▷	低压异常延时 : 5 秒	
Flushing method	Low pressure anomaly delay	
持续制水时长 : 4 时	启动制水延时 : 1 分	
Continuous water production time	Start water-making delay	
低压检测方式 : 持续检测 ▷	低压检测延时 : 10 秒	
Low pressure detection method	Low pressure detection delay	
保存	恢复出厂	退出
Save	Restore to factory	Exit

Parameter name	Factory default	Set the range	Meaning
运行冲洗时间 Run time flush	15 Second	0-240	Rinse duration during water production

运行冲洗间隔 Run wash interval	60 Minute	1-240	Water-making process flush interval
待机冲洗时间 Standby flush time	15 Second	0-100	Pure water tank water full flush duration
待机冲洗间隔 Standby flush interval	2 hours	1-12	Pure water tank water full flush interval time
冲洗方式 Flushing method	高压冲洗 High pressure flush	高压/低压 冲洗 High pressure/low pressure flush	Reverse osmosis membrane flush mode
低压异常延时 Low pressure anomaly delay	5 Second	1-30	Low voltage switch abnormal waiting time, this time is processed according to the normal low voltage switch
持续制水时长 Continuous water production time	4 hours	0-12	Maximum continuous running time of high pressure pump
启动制水延时 Start water-making delay	1 minute	0-30	Pure water tank below high level delay water time

低压检测方式 Low pressure detection method	持续检测 Continuous detection	持续/启动 检测 Continuous/st art detection	Whether the low pressure abnormality is judged in the water production process. When starting detection, only judge whether the low pressure is normal before starting water production. Low abnormality during water production does not affect water production.
低压检测延时 Low pressure detection delay	10 Second	1-30	Wait for the normal time of the low pressure switch after the raw water pump is started and before the high pressure pump is started when the water production is started, and alarm and switch to the fault handling mode corresponding to the operation state after continuously detecting 5 times of low pressure abnormality.

After modification, click the  button to save the parameters, and click the exit button to not save the modified

value;

Click the **恢复出厂** button to automatically reset the run parameters and system parameters to the factory default values;

System parameter setting

- 1. Tap the **系统参数** System Parameters button on the main interface to enter the settings interface;
- 2. The range and description of the system parameters are shown in the following figure;

杀 菌 时 间 : 30 秒	杀 菌 间 隔 : 2 时	
Disinfection time	Disinfection interval	
加 热 启 动 温 度 : 5 °C	加 热 停 止 温 度 : 10 °C	
Heating start temperature	Heating stop temperature	
纯 水 水 质 校 准 : 0 us/cm	原 水 水 质 校 准 : 0 us/cm	
Pure water quality calibration	Source water quality calibration	
异 常 启 动 延 时 : 5 分	息 屏 时 间 : 5 分	
Abnormal start delay	Screen off time	
屏 幕 亮 度 : <input type="range" value="50"/>		
Screen brightness		
保存	厂商信息	退出
Save	Manufacturer information	Exit

参数	出厂默认	范围	含义
杀菌时间 Disinfection time	30 Second	1-240	Ozone generator continuous sterilization time

杀菌间隔 Disinfection interval	2 hours	1-24	Ozone generator start interval time
加热启动温度 Heating start temperature	5℃	0-15	Heat preservation heating start temperature
加热停止温度 Heating stop temperature	10℃	0-15	Heating temperature cutoff, heating does not start when the temperature is less than the heating start temperature after heating stops
纯水水质校准 Pure water quality calibration	0 uc/cm	-50-100	and pure water detection value accumulation
原水水质校准 Source water quality calibration	0 uc/cm	-50-1000	and the original water detection value accumulation
异常启动延时 Abnormal start delay	5 Minute	1-30	Low pressure abnormality, water production timeout or water replenishment timeout occurs during operation, the equipment suspends and then starts after a delay.
息屏时间 Screen off time	5 Minute	0-10	Screen activation display brightness duration, switch to screen saver

			brightness after timeout (fixed brightness 10), do not switch screen saver brightness after
屏幕亮度 Screen brightness	100	5-100	Brightness when screen is active

After modification, click the  button to save the parameters, and click the

 button to not save the modified value;

Click the  button to jump to the product information interface;

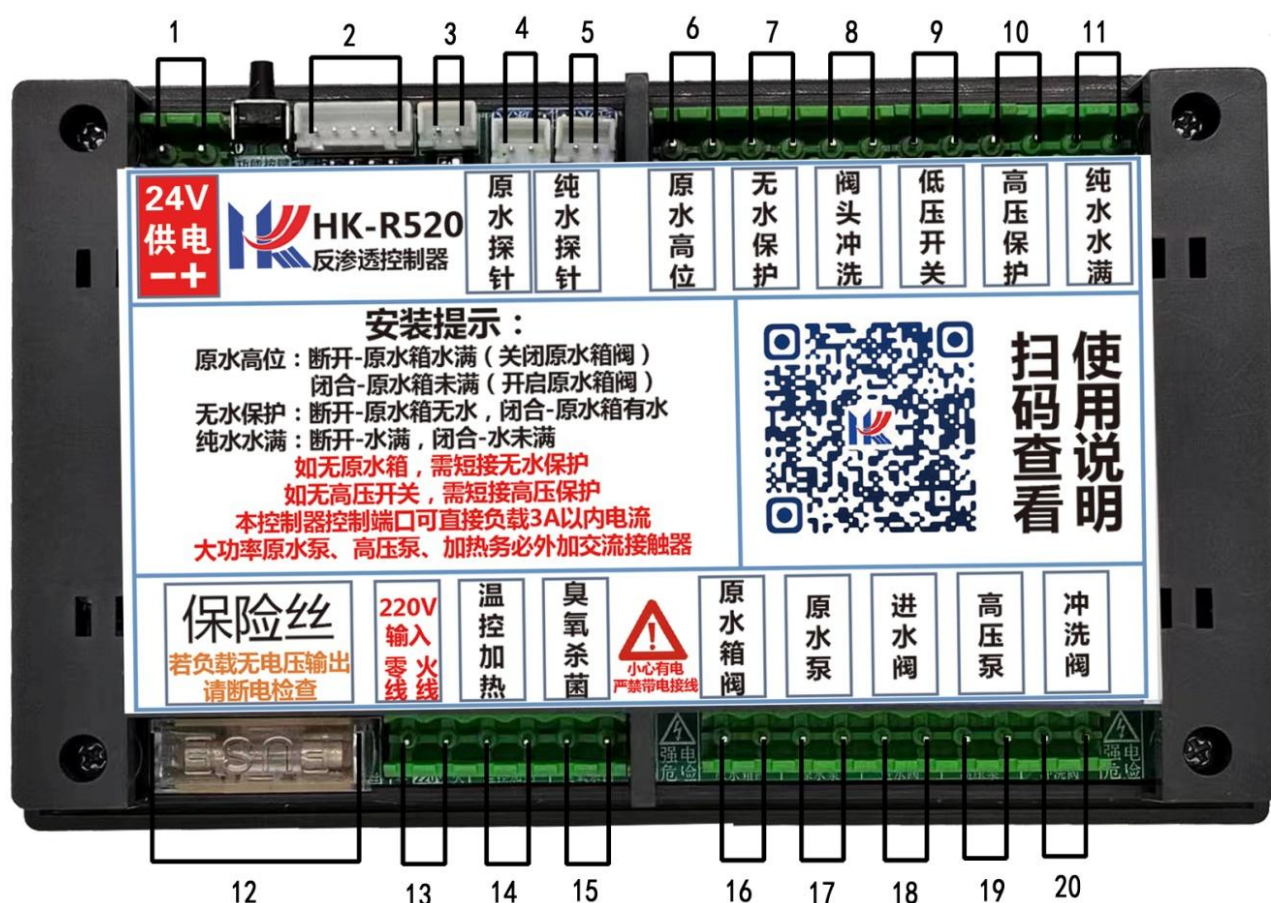
Default process

1. Enter the setting parameters interface, if no operation is performed within 3 minutes, the system will automatically exit and jump to the running interface
2. Long press the function key for 2 seconds to restart the display, and pressing the key within 10 seconds after releasing it will be invalid;
3. Without connecting the temperature probe, the operation interface does not display the temperature value. When the insulation heating is started, it will automatically turn off for 1 minute and then turn on again if it is continuously heated for 10 minutes. When the set start temperature value is less than the stop temperature, the insulation heating will not.
4. The equipment suspends after low pressure abnormality and

water-making overtime, and the equipment automatically runs after the abnormality startup delay time;

5. The feed water tank valve will automatically close after being kept open for 4 hours, and will reopen after waiting for the abnormal start-up delay time

Connection port description



1. 24V power supply: motherboard power supply;
2. Standby port: 5P port, spare for special functions;
3. Temperature probe: Temperature detection port, a

temperature probe needs to be connected when insulation control is required;

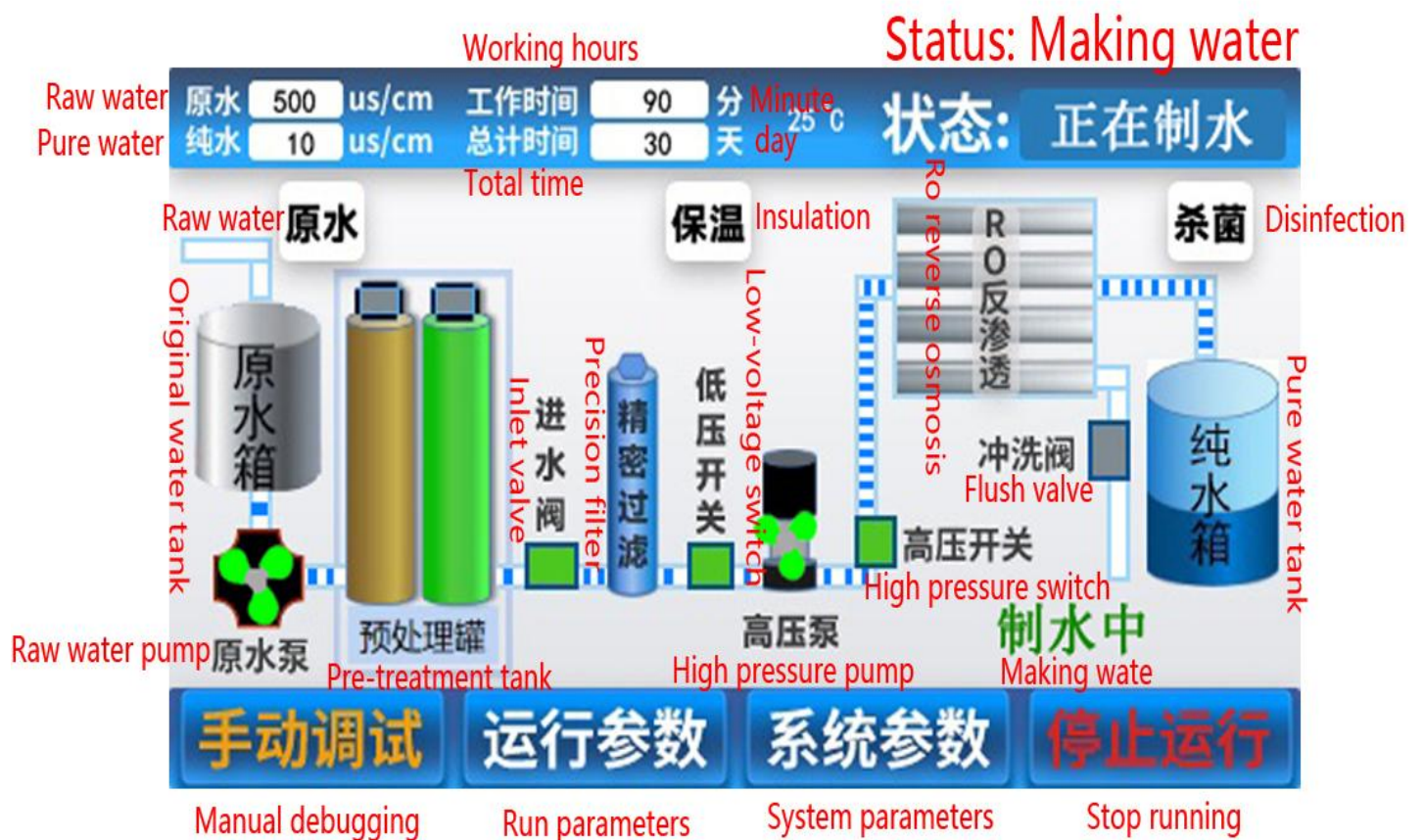
4. Raw water probe: raw water conductivity detection port;
5. Pure water probe: pure water conductivity detection port;
6. Raw water high level: High water level detection of raw water tank, open the valve of raw water tank when the tank is full of water, close the when the tank is not full of water, and open the valve of raw water tank after closing.
7. No water protection: Low water level detection of the original water tank, water tank without water disconnection, with water closure, closure can only start water, (if there is no original water tank, this port needs to be short-circuited);
8. Valve head flushing: Self-flushing valve head detection, self-flushing is disconnected when closed, closed when started, and the raw water pump started for self-flushing operation when the raw water tank has water after closing;

9. Low-voltage switch: Water pressure detection before the high-pressure pump, open when there is no water pressure, close when there is water pressure, and high-pressure pump can only be started when it is closed;
10. High pressure protection: High water pressure detection before the reverse osmosis membrane, the water pressure is high to disconnect, the water pressure is low close, stop making water when disconnected, to prevent high water pressure from damaging the reverse osmosis membrane, (if there is no high pressure switch, this port needs to short-circuited);
11. Pure water full: High water level detection of pure water tank, water full disconnection, water not full closing, closing is required to start production;
12. Fuse: Use a 10A fuse by default;
13. 220V input: Control end power input, wiring sequence left

zero right fire, control port output voltage is 220V

14. Temperature control heating: Temperature control output, a temperature control probe is required to be connected, and the temperature control will not start without connecting a temperature control;
15. Ozone sterilization: Ozone sterilization control output, automatically operate according to the set sterilization time and sterilization interval
16. Raw water tank valve: The raw water tank valve controls the output, starting when the raw water is at a high level;
17. Raw water pump: The raw water pump control output, external AC contactor controls the start of the raw water pump;
18. Inlet valve: The inlet valve controls the output, starting when the raw water pump is started and there is no self-flushing

19. High pressure pump: The high pressure pump controls the output, and an external AC contactor controls the start of the high pressure pump;
20. Flush valve: Flush valve controls output, starts during flushing;





安装提示:

原水高位: 断开-原水箱水满 (关闭原水箱阀)

闭合-原水箱未滿 (开启原水箱阀)

无水保护: 断开-原水箱无水, 闭合-原水箱有水

纯水水满: 断开-水满, 闭合-水未滿

如无原水箱, 需短接无水保护

如无高压开关, 需短接高压保护

本控制器控制端口可直接负载3A以内电流

大功率原水泵、高压泵、加热务必外加交流接触器

Installation tips

High level of raw water: Disconnect - Raw water tank is full(Close the raw water tank valve)

Close Raw water tank is not full (Open the raw water tank valve)

No water protection: Disconnect - No water in raw water tank, Close - Water in raw water

Pure water full: Disconnect - Full, Close - Not full

If there is no original water tank, the no-water protection needs to be short-circuited

If there is no high-pressure switch, the highpressure protection needs to be short-circuited

The control port of this controller can directly load a current of 3A or less

Large power original water pump, highpressure pump, heating must be added with an AC contactor